

Bonian Han

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Education

Hangzhou Dianzi University *BS in Statistics*

Sept 2021 - July 2025

- GPA: 3.48(84.8/95)

New Jersey Institute of Technology *PhD in Computer Science*

Sept 2025 - May 2026

- GPA: 4.0

- Advisor: [Prof. Zhi Wei](#) 

Boise State University *PhD in Computer Science*

Sept 2026 - present

- GPA: 4.0

- Advisor: [Prof. Yu Zhang](#) 

Skills & Interests

Research Interests: Machine Learning, Computer Vision, AI Agent, Large Language Model

Skills: Python, C/C++, PyTorch, SQL, R, Linux, Bash, Git, Docker

Languages: TOEFL: 91, GRE: 326 (Verbal: 156, 69th percentile; Quantitative: 170, 92nd)

Publications

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1. J. Zhang, **B. Han**, G. Liang, and Y. Zhang. (2026). Rethinking Learned Occupancy in Autonomous Active Mapping with Observation-Gated Filtering. *IEEE/RSJ IROS 2026 Workshop on Space Exploration and Sustained Operations Beyond Earth*. **Oral; Best Paper Runner-Up Award**.
 2. **Han, B.**, Qi, C., Musialski, P., and Wei, Z. (2026). INST-Align: Implicit Neural Alignment for Spatial Transcriptomics via Canonical Expression Fields. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. Accepted. [Paper](#) .
 3. **Han, B.**, Masupalli, Y. P., Xing, X., and Liang, G. (2024, December). Improving Medical Imaging Model Calibration through Probabilistic Embedding. *2024 IEEE International Conference on Big Data (IEEE BigData 2024)*. DOI:10.1109/BigData62323.2024.10825661 .
 4. **Han, B.**, Moran, C., Yang, J., Lee, Y., Cao, Z., and Liang, G. (2024, July). Multi-Scale Self-Supervised Consistency Training for Trustworthy Medical Imaging Classification. In *Proceedings of the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. Podium (acceptance rate approx. 28%). H5-Index: 48. DOI: 10.1109/EMBC53108.2024.10782322 .
 5. Zulu, J., **Han, B.**, Alsmadi, I., and Liang, G. (2024, March). Contextualized Embedding-Based Approach for Enhanced SQL Injection Detection. In *Proceedings of the 62nd ACM SouthEast Annual Conference*. DOI:10.1145/3603287.3651187 .

Experience

- Research** with Dr. Gongbo Liang, Texas A&M University-San Antonio Nov 2023 - Nov 2024
- Conducted research on computer vision, focusing on model calibration and feature extraction to reduce model bias. Three papers accepted: *IEEE EMBC*, *ACMSE*, and *IEEE Big Data*.
- Research** with Prof. Zhi Wei, New Jersey Institute of Technology Sept 2025 - May 2026
- Conducted research on spatial transcriptomics and neural representation methods. Worked on INST-Align, coupling spatial alignment with a canonical expression field; accepted to MICCAI 2026.
- Working** Ballkids Coach & Manager Sept 2024
- Trained and managed professional ballkids (ages 9-14) at the Hangzhou Open ATP 250 and ATP Challenger, ensuring smooth match operations, coordination of on-court schedules across departments, and adherence to court etiquette and safety protocols.
- Volunteer** 19th Asian Games and Asian Para Games Sept 2023 - Oct 2023
- Served as a Tennis Ballkids & Ceremony Assistant, managing on-court operations and assisting players and umpires; featured by China Central Television (CCTV) and honored as an Outstanding and Excellent Volunteer.

Awards & Honors

- Best Paper Runner-Up Award**, IEEE/RSJ IROS 2026 Workshop Sept 2026
- Outstanding & Excellent Volunteer**, National 2023
- Second Prize**, Statistics Research Competition, Provincial (State) 2023
- Third Prize**, National College Students Mathematics Competition (Online Challenge) 2022
- Fifth Place**, University Basketball Annual Competition May 2022